KNUST **RANKS NO.1 GLOBALLY** FOR THE PROVISION OF QUALITY EDUCATION **(SDG 4)** By: THE WORLD UNIVERSITY RANKINGS

Welcome to “Digital Signal Processing”

SDG 4 QUALITY EDUCATION

KNOW YOUR LECTURER

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- Research Interests: VANETs, Artificial Intelligence, Smart Systems, Massive MIMO, Wireless and Mobile Communication, IoT, BI

Introduction

- Basic concepts and terminology in DSP

- Comparison between analog and digital signal processing.

- Description of discrete signals and systems in time domain

Introduction

- Description of discrete signals and systems in frequency domain
- Discrete Fourier Transform (DFT) and inverse transform properties
- Circuit convolution and its relation to DFT, Fast Fourier Transform (FFT)

Introduction

Areas of Application of DSP:

Image and voice processing

Telecommunications

Industrial Control

Radar and Sonar

Biomedical Analysis.

Related Courses

- "Signals and Systems", very important!
- "Advanced Mathematics", "Linear Algebra"
- "Programming Language"
- "The Principle of Micro-computer"
- "Logic Circuit"
- "The Technique of DSP"

Course Assessment

- Exams - 50%
- Project – 20%
- Homework - 20%
- Unannounced Quizzes – 10%
- Tests will be organized at the end of every topic
- Slides will be released at the end of the last slide for the topic
- Student should note key points during lectures.

What is DSP?

Digital Signal Processing : — Theories, methods and algorithms about how to process signals in digital form.

Digital Signal Processor: — A kind of microprocessor used to implement digital signal processing algorithms;

— Architecture (Chips, Peripherals, Pipelines, Instructions, Circuits, … …)

— System (Hardware & Software Design and Debug … …)

What is DSP?

- DSP simply means processing signals in digital form.
- Analog v.s. Digital
  - Analog → Continuous in both time and magnitude
  - Digital → Discrete in both time and magnitude
  - Most signals in real world are analog!
  - Needs some transformations when performing DSP
    - Step 1: Sample the analog signal at some discrete time points
    - Step 2: Quantize the sample values either by rounding or truncation

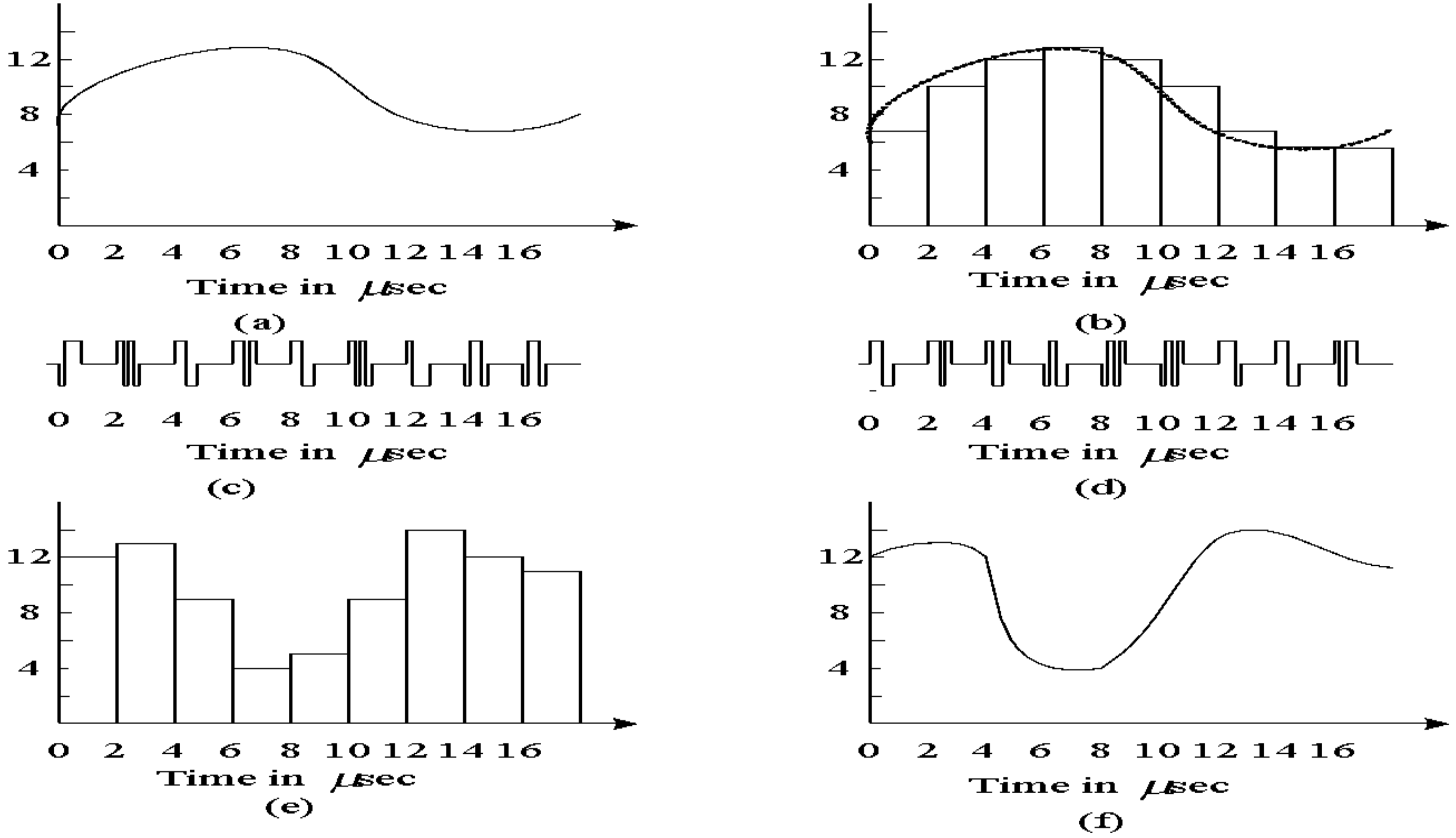
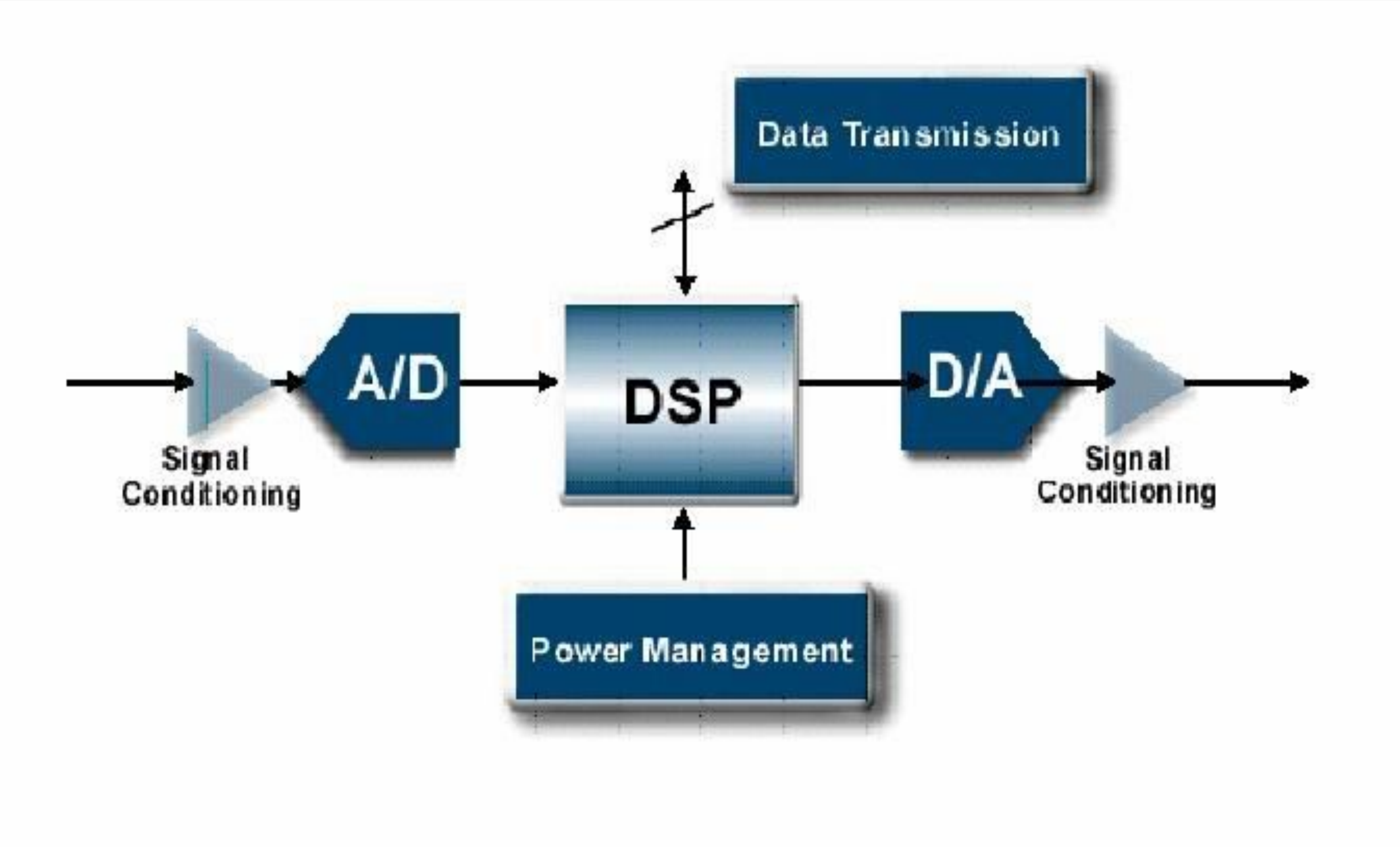


Fig 1.43: Typical waveforms of signals

DSP Solution



Why DSP is important?

- The foundation of information technology is digitization.
- The kernel of digitization is digital signal processing.
- Most digital signal processing, especially real-time processing are implemented by DSP processors or Application Specific Integrated Circuits(ASICs) based on DSP cores.
- DSP technology has become a hot front edge and driver for the entire semiconductor industry.

Why Digital?

- Advantages over ASP:
  - More reliable, less sensitive to tolerances such as temperature, noise, etc.;
  - Higher accuracy, can be integrated on a single chip;

- Limitation:
  - A/D D/A rates are not available in some applications;(beyond GHz) and (a few Hz)

Why Digital?

(1) Programmability:

- Analog systems: You need to modify hardware design
- Digital systems: Only modify software

Examples:

Analog filter, digital filter, adaptive filter, … …

Why Digital?

(2) Precision:

- Analog system by components specification.eg. Resistors have a tolerance of 5%, Capacitors 20% or worse;
- Digital system by ADC bits, CPU word width (or word-length) and algorithm

Why Digital?

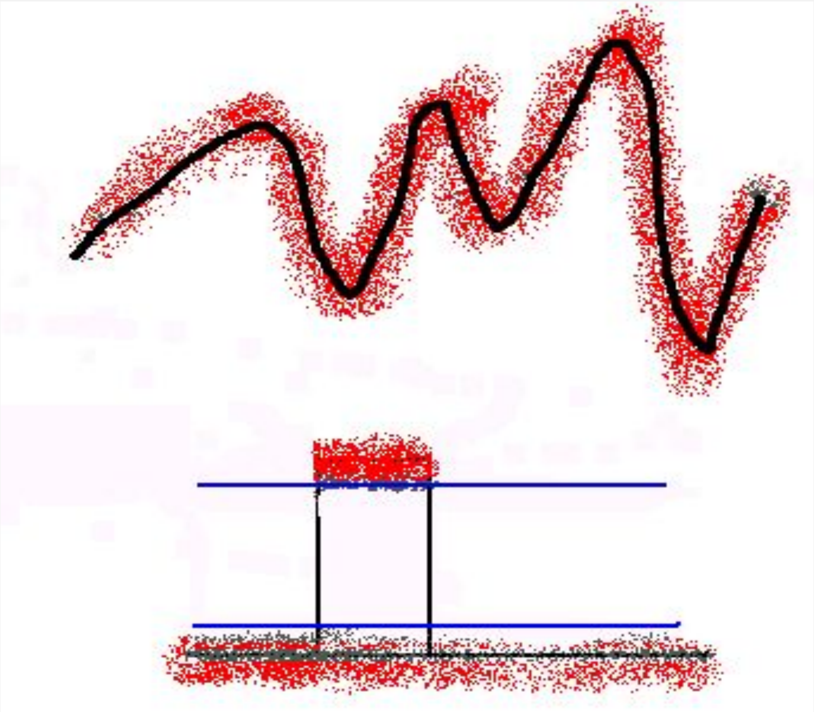
(3) Stability:

- Analog system: The characteristics of analog system components (e.g., resistors, capacitors and operational amplifiers) will change along with temperature, humidity, etc.

- Digital system: Shows no variation with temperature throughout their guaranteed operation range.

Why Digital?

(4) Anti-noise:



Why Digital?

(5) Repeatability:

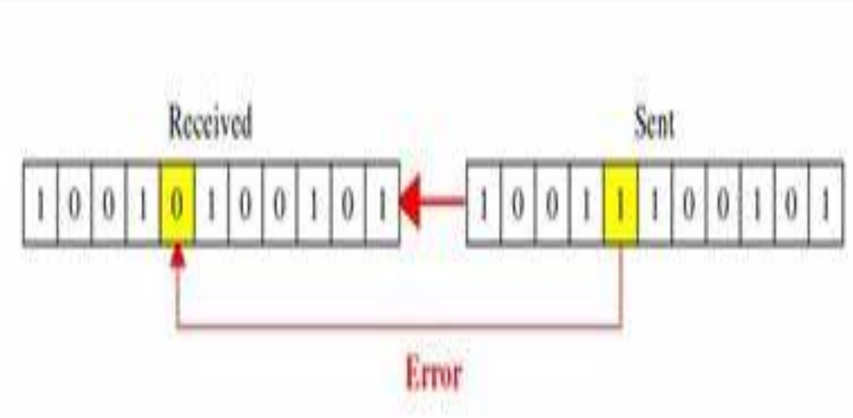
Why Digital?

(6) VLSI (Very Large Scale Integrated Circuit):

Why Digital?

(7) Error Correcting Codes:

- Data retrieval and transmission systems suffer from a number of potential forms of error. With information in a digital or binary form, we may easily build into the data stream additional "**redundant bits**" that are used to detect when an error has occurred.
- An error-correcting code is an algorithm for expressing a sequence of numbers such that any errors which are introduced can be detected and corrected .



Why Digital?

(8) Data Transmission and Storage:

- The fidelity of the digital medium is greater than that of the analog one.
- The Internet, Compact Disc (CD) and Digital Video Disc (DVD) brought information based on trouble-free high-quality text, audio and video into offices and homes.

Why Digital?

(9) Data Compression:

The information channels cost and transmission bottlenecks makes compression necessary for real-time processing.

With analog compression some information is lost.

An example is the bandwidth limiting applied to analog telephone lines, which limits the bandwidth to 3kHz.

Digital tech. makes lossless compression possible.

We still need analog processing!

(1) Real-Time Processing:

- Analog system :
  - Besides the delay introduced by the circuit, the processing is in real-time.
- Digital system :
  - Processing time is determined by the processor speed and the adopted algorithms.

We still need analog processing!

(2) Processing very high frequency signals:

Analog system :

Can process microwave, milli-meter-wave, even light wave signals.

Digital system :

By the Nyquist Rule, the processing is limited by the S/H, A/D and processor speed.

ADs with very high sampling rates are extremely expensive!

We still need analog processing!

(3) Most signals in real world are analog

If we want to process these analog signals with a digital signal processing system, we must change them into digital form first by **mixed signal processing**.

APPLICATIONS OF DSP

1. COMMUNICATION

- **Wireless Communication: Base stations, Switching Centers and Mobile phones.**

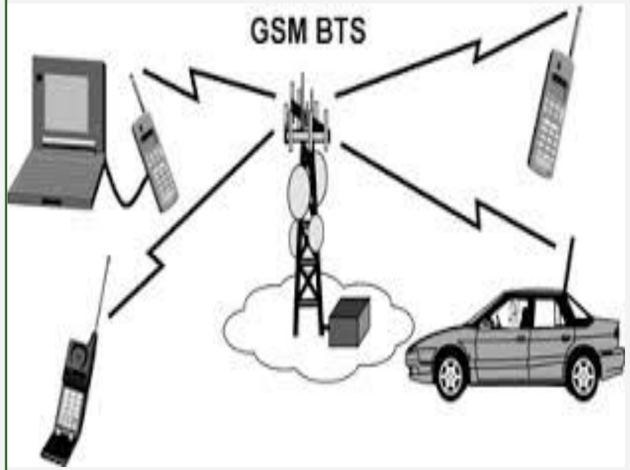
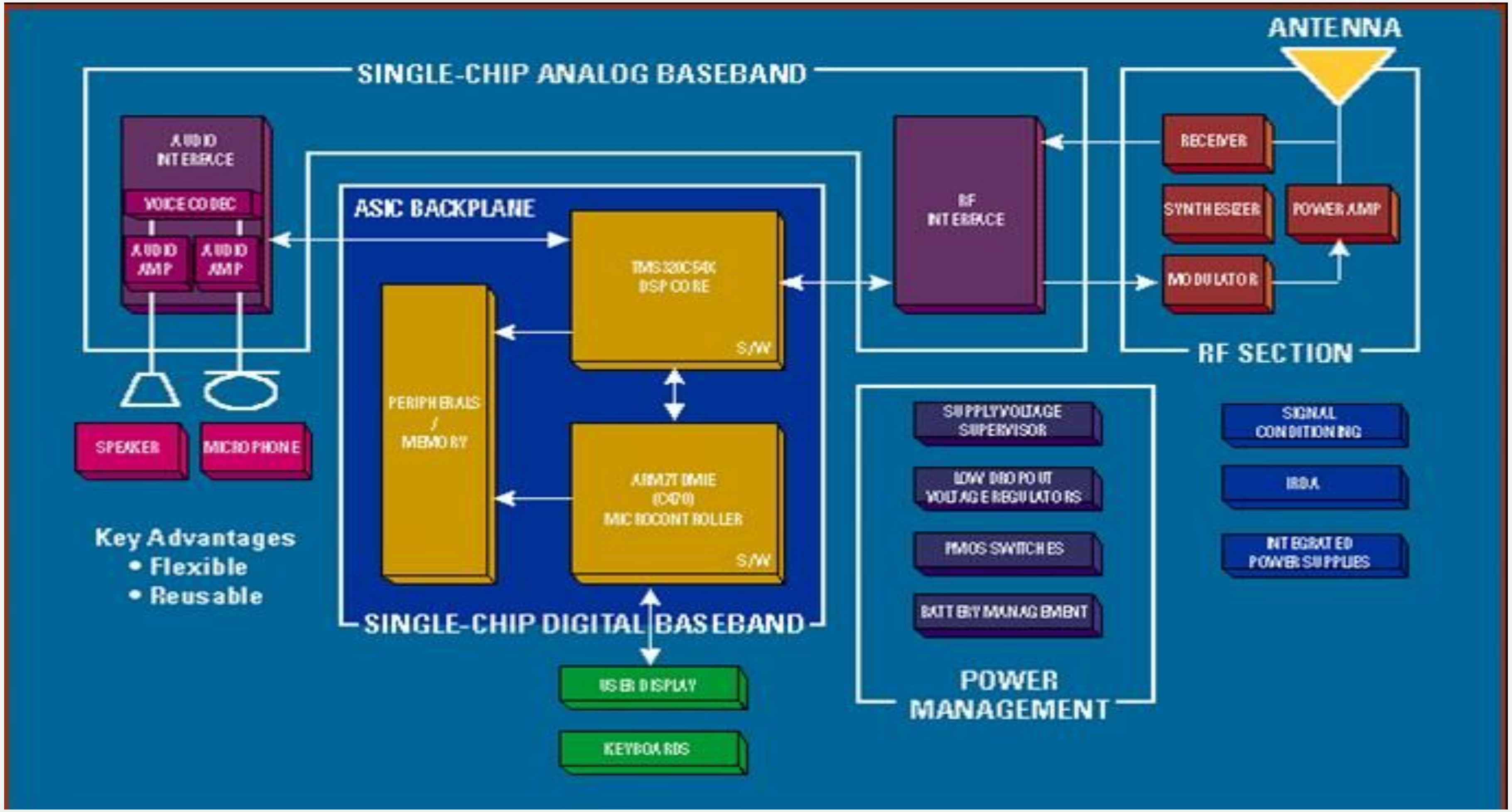
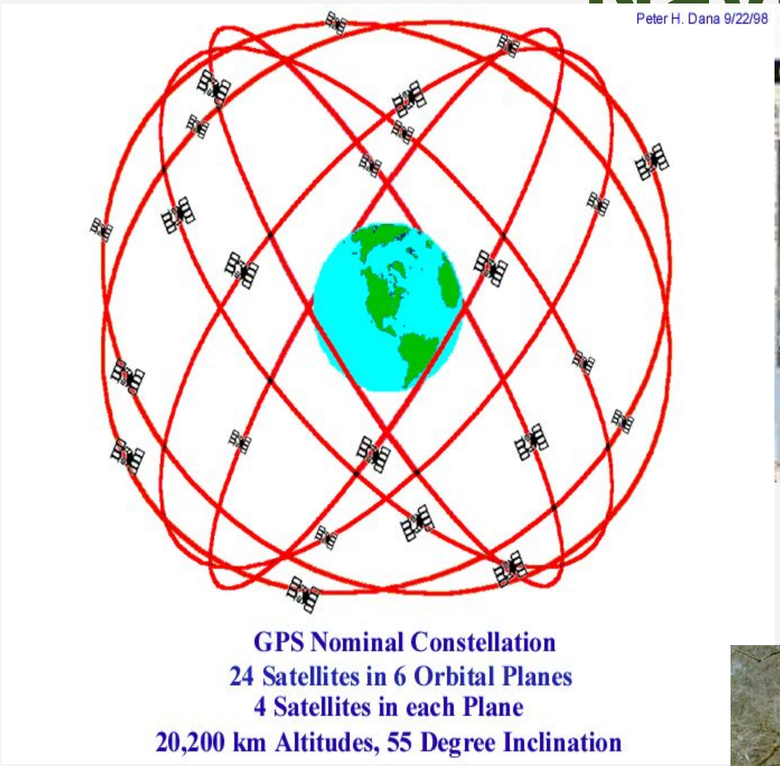


Diagram of GSM Cell-Phone



2. Satellite Navigation

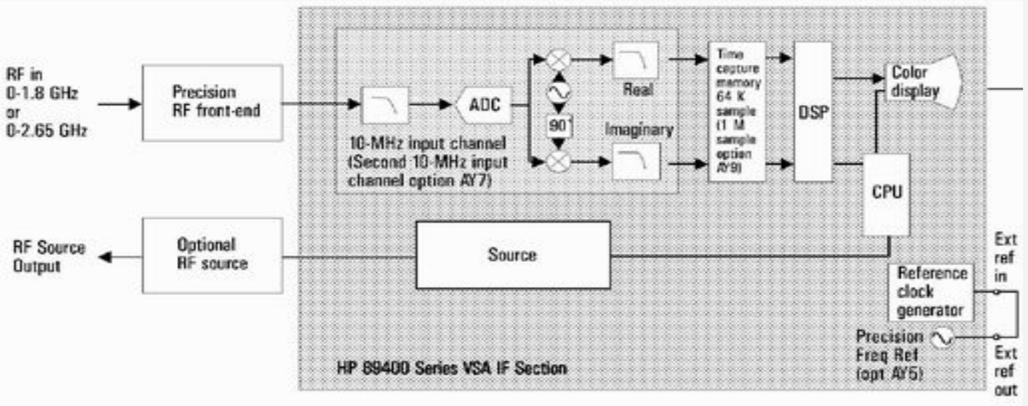


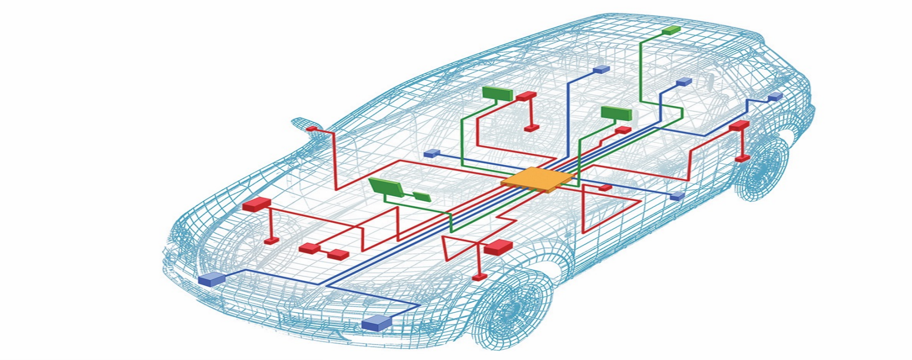
- GIS
(Geographic Information System)

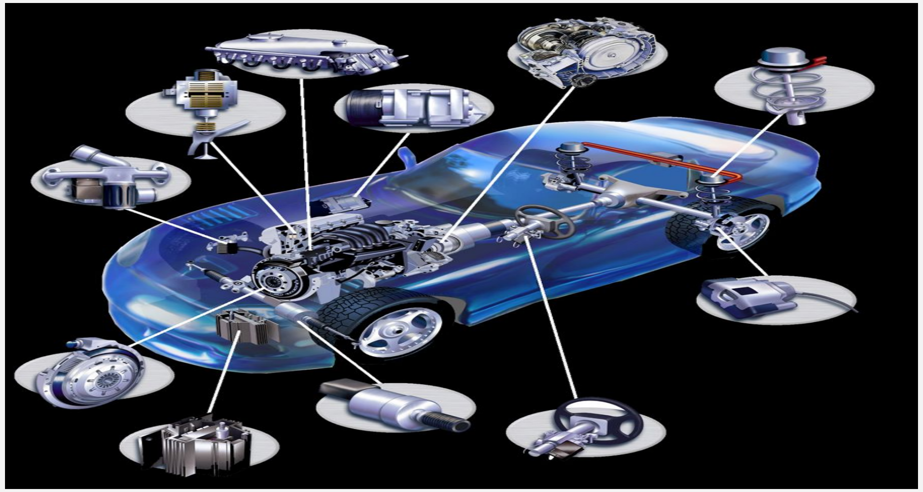
- GPS
(Global Positioning System)

3. Measurement and Control

- Virtual Instruments (HP89441 Vector Analyzer)







- Automotive Electronics

4.Military and Safety

Air Early Warning (AEW)

Radar/Sonar

Cruise Missile

4.Military and Safety

Smart Bomb

UAV (Unmanned Aerial Vehicle)

5. Consumer Electronics

DV (Digital Video)

MP3 (MPEG Level-3 Audio Decoder)

DC (Digital Camera)

5. Consumer Electronics

HDTV (High Definition TV)

IP Phone

Home Theater

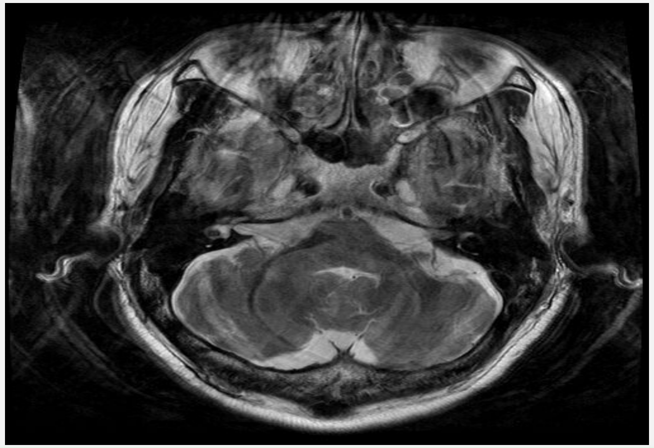
IPTV

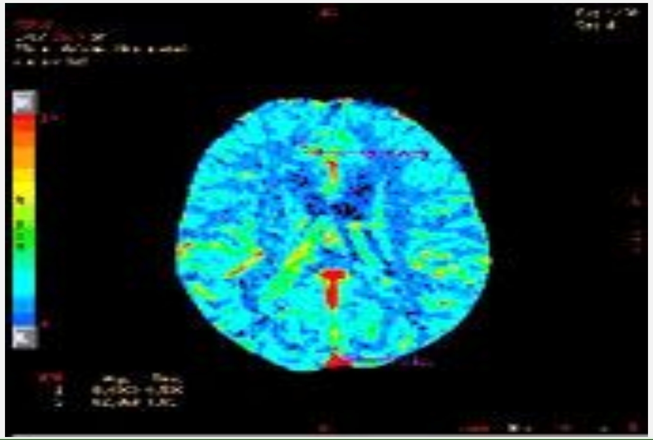
6. Medical

Ultrasound

MRI
(Magnetic Resonance Imaging)

CT (Computed Tomography)

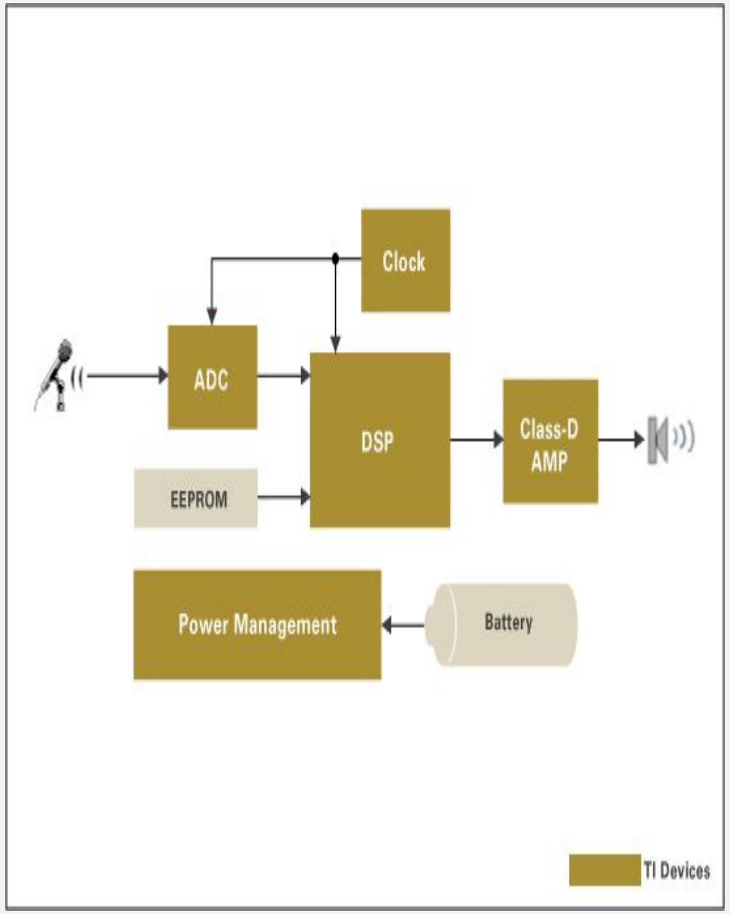




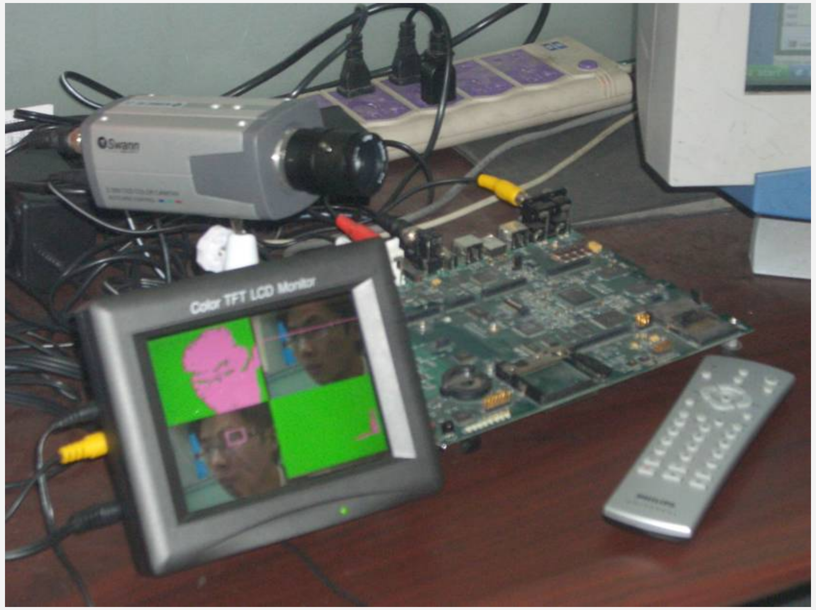
6. Medical

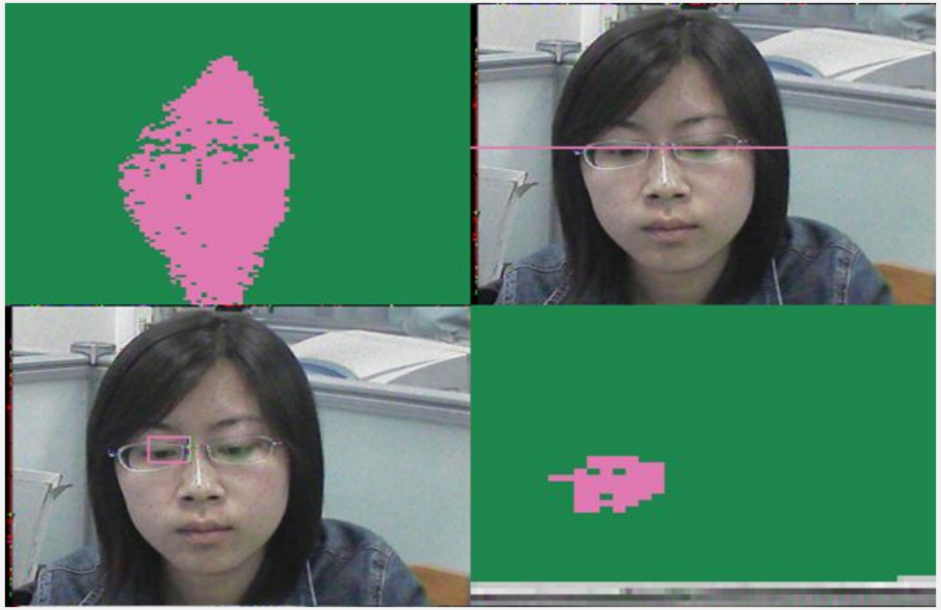
Gamma knife

Hearing Aids



Drowsy Driving Alert System

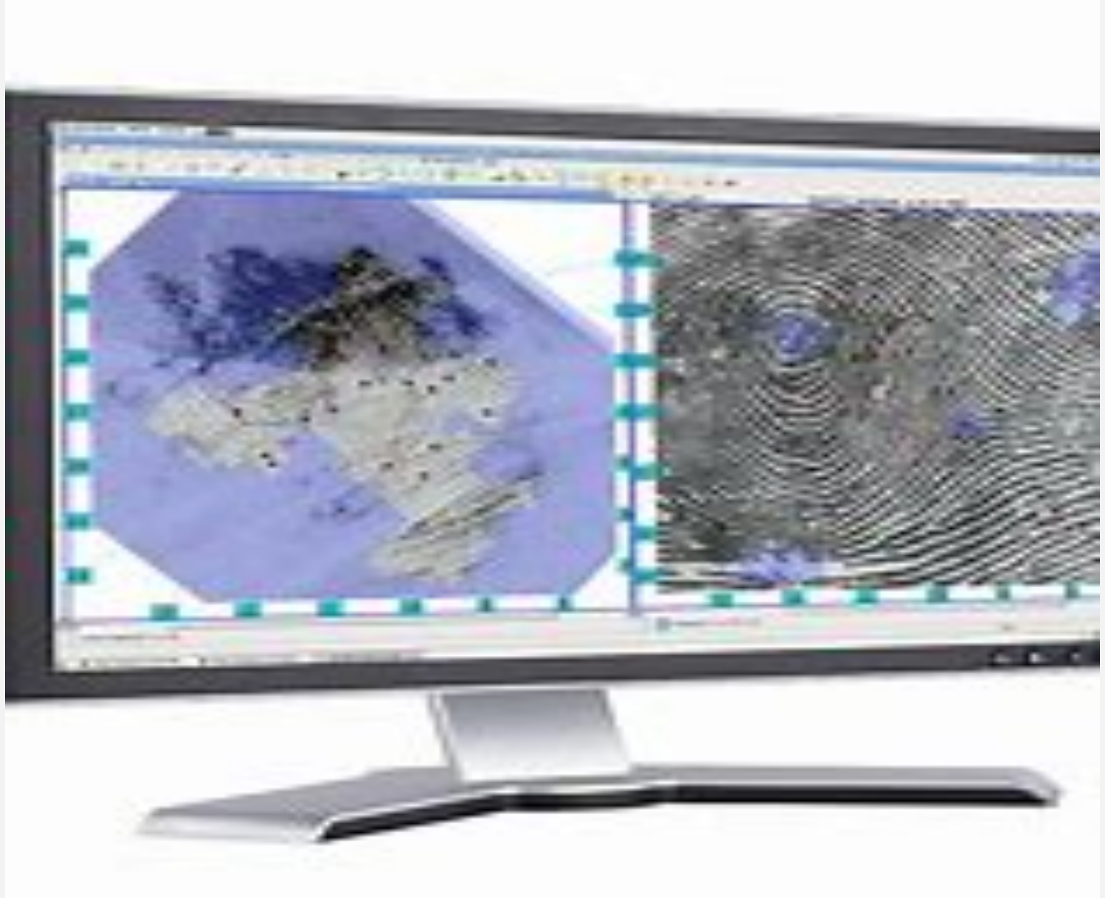




Digital Watermarking

Video Surveillance System

Fingerprint Identification



Pattern Recognition

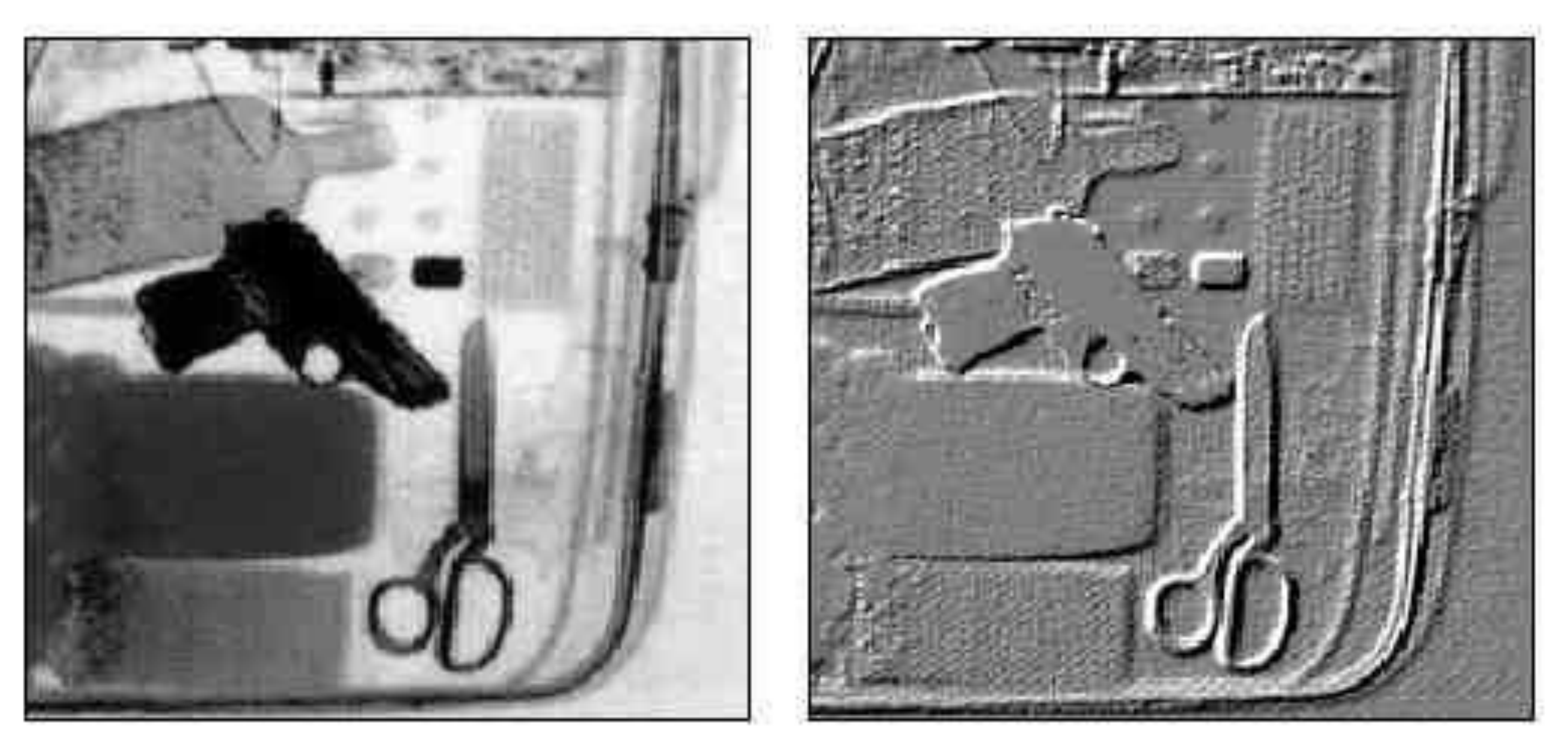
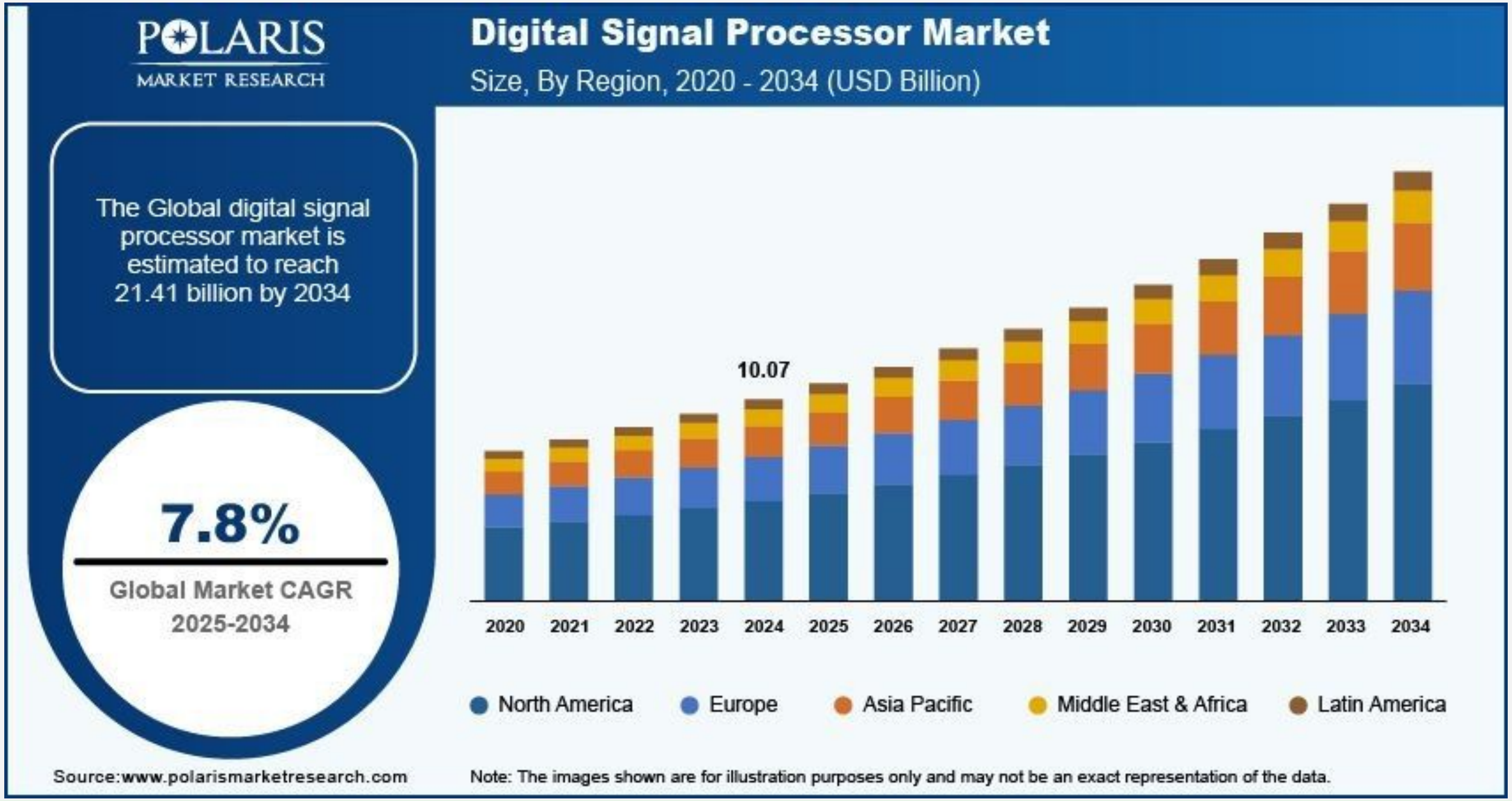


Image & Video Reparation

DSPS MARKET WILL EXPLODE!



Home Work 1

Pick any area of your choice and submit a one page paper(double column) on how DSP has been innovatively applied to bring about major changes in the area. You must cite at least 10 credible references.

Home Work 2

Select any of the DSP applications discussed and do a technical presentation of how DSP has been implemented in the industry.

Short test in the next class!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